

AI-Based Log Aggregation Techniques for SIEM

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CSP650



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Project Background



- Organizations use various devices and systems, such as firewalls, Android devices, and IoT devices, that generate security logs.
- These logs contain valuable information for security monitoring and threat analysis.
- SIEM systems collect and analyze logs from multiple sources to help security analysts monitor security events.
- The use of **AI and Machine Learning** can help **automate** the analysis and classification of large volumes of security logs.

Problem Statement



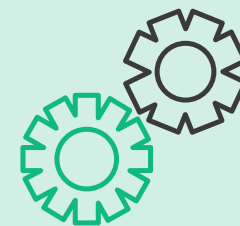
- Logs generated by **different environments** often have **different formats and structures**.
- The differences in log formats make it **difficult** to combine, process, and analyze logs efficiently.
- Security analysts may face **challenges in identifying and prioritizing potential threats** from large volumes of heterogeneous logs.
- Therefore, an **AI-based** log aggregation approach is needed to standardize logs and support **automated** security event classification and risk assessment.



Objectives



1. To **design** log aggregation techniques for SIEM using AI-based approaches.



2. To **develop** an AI-Based Log Aggregation Techniques for SIEM that unifies diverse log formats from multiple sources into one dataset.



3. To **evaluate** the effectiveness of AI-Based Log Aggregation Techniques in improving SIEM systems.

Scope & Significance



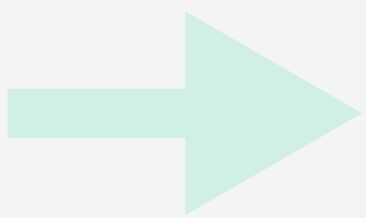
Scope

- Collects and processes logs from **Firewall, Android, and IoT** sources.
- **Unifies logs** into a common SIEM format.
- Applies **NLP and ML** for log classification and risk scoring.
- Displays results through a SIEM dashboard.

Significance

- **Simplifies** the management of heterogeneous security logs.
- Supports **automated** attack detection and risk assessment.
- Helps security analysts **improve security monitoring** and threat analysis **across different environments**.

Literature Review



Author & Paper	Problem Addressed	Solution Proposed	Technology Used
Saleem Raja, Vinodini, & Kavitha (2021) – <i>Lexical features based malicious URL detection using machine learning techniques</i>	Detecting malicious requests using only lightweight lexical text features, without deeper content or network-level analysis	Extracts lexical features from raw URL/request text and trains a Random Forest classifier to separate malicious from legitimate requests	Random Forest, supervised learning, lexical text feature extraction
Abdelaziz et al. (2024) – <i>Enhancing network threat detection with random forest-based NIDS and permutation feature importance</i>	Improving network intrusion detection accuracy while retaining interpretability	Trains a Random Forest classifier on network traffic features and uses permutation feature importance to rank the most discriminative attributes	Random Forest, supervised learning, permutation feature importance
Azizi Doost et al. (2025) – <i>A new intrusion detection method using ensemble classification and feature selection</i>	Reducing noise and dimensionality in intrusion detection while maintaining high accuracy	Combines automated feature extraction with a Random Forest classifier to achieve high accuracy and precision on benchmark intrusion datasets	Feature extraction, Random Forest, ensemble classification
Almotairi, Atawneh, Khashan, & Khafajah (2024) – <i>Enhancing intrusion detection in IoT networks using machine learning-based feature selection and ensemble models</i>	Improving intrusion detection performance on heterogeneous IoT network traffic	Applies feature selection together with a heterogeneous ensemble classifier combining multiple base learners	Feature selection, heterogeneous ensemble learning, IoT network traffic
Kalakoti, Vaarandi, Bahsi, & Nömm (2025) – <i>Evaluating explainable AI for deep learning-based network intrusion detection system alert classification</i>	Improving analyst trust and interpretability of AI-generated intrusion alerts	Evaluates post-hoc explainability techniques for network intrusion detection alert classification and analyst decision support	Explainable AI, alert classification, analyst-facing interpretability
<i>AI-Based Log Aggregation Techniques for SIEM</i>	Security logs from Firewall, Android, and IoT sources have different formats and structures, making them difficult to combine, process, and analyze efficiently.	Develop an AI-based log aggregation framework that standardizes and unifies logs from multiple sources, then classifies security events and assesses their risk to support SIEM monitoring.	NLP, Feature Engineering, ML (Random Forest), SIEM Dashboard

Methodology & Flowchart

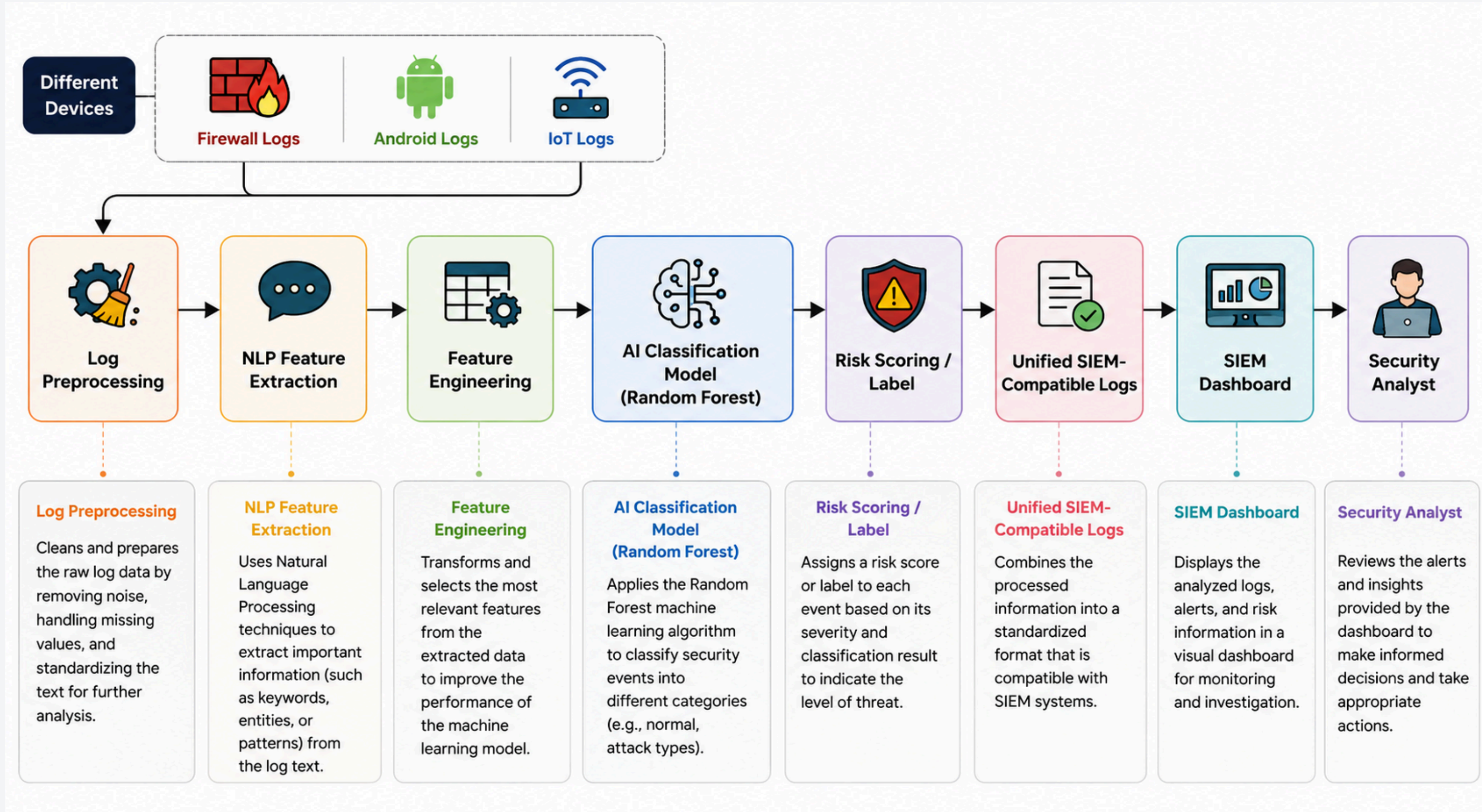


The project follows the Agile methodology consisting of:

1. Requirements
2. Design
3. Development
4. Testing
5. Deployment
6. Review



Flowchart



Results

	Rows	% of Dataset	Notes
Training set	75, 329	80.00%	Model learns from these (sees data + correct answer)
Total set	18, 8333	20.00%	Model is evaluated on these — never seen during training
Total dataset	94, 162	100%	Combined — used later for full-dataset scoring only

Metric	Measured on	Rows	Correct predictions	Result
Accuracy	Test set only (unseen data)	18,833	15,030	79.80%
Agreement	Full dataset (train + test combined)	94,162	90,034	95.60%

Results

Metric	Value	Notes
Accuracy	79.81%	Overall held-out test-set accuracy (18,833 events)
ROC-AUC	89.76%	Area under the ROC curve for the attack class
Attack precision / recall / F1	73.88% / 74.29% / 74.09%	Class 'attack', support = 7,317
Normal precision / recall / F1	83.61% / 83.31% / 83.46%	Class 'normal', support = 11,516
Macro-average F1	78.77%	Unweighted average across both classes

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Accuracy: 0.7980672224287155

Classification Report:
              precision    recall  f1-score   support

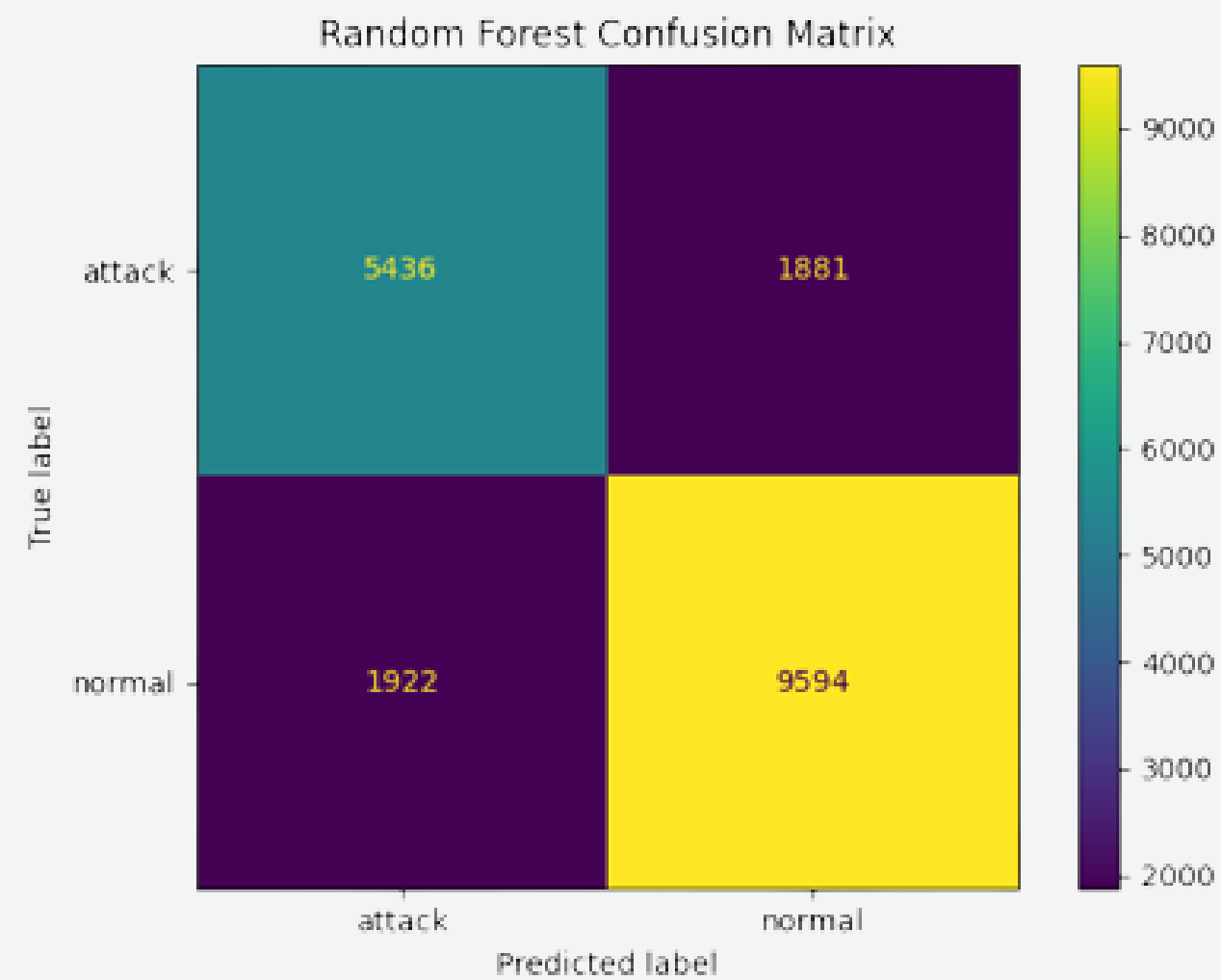
   attack       0.74       0.74       0.74       7317
   normal       0.84       0.83       0.83      11516

   accuracy                   0.80      18833
  macro avg       0.79       0.79       0.79      18833
 weighted avg     0.80       0.80       0.80      18833

Confusion Matrix (rows/cols ordered [attack, normal]):
[[5436 1881]
 [1922 9594]]

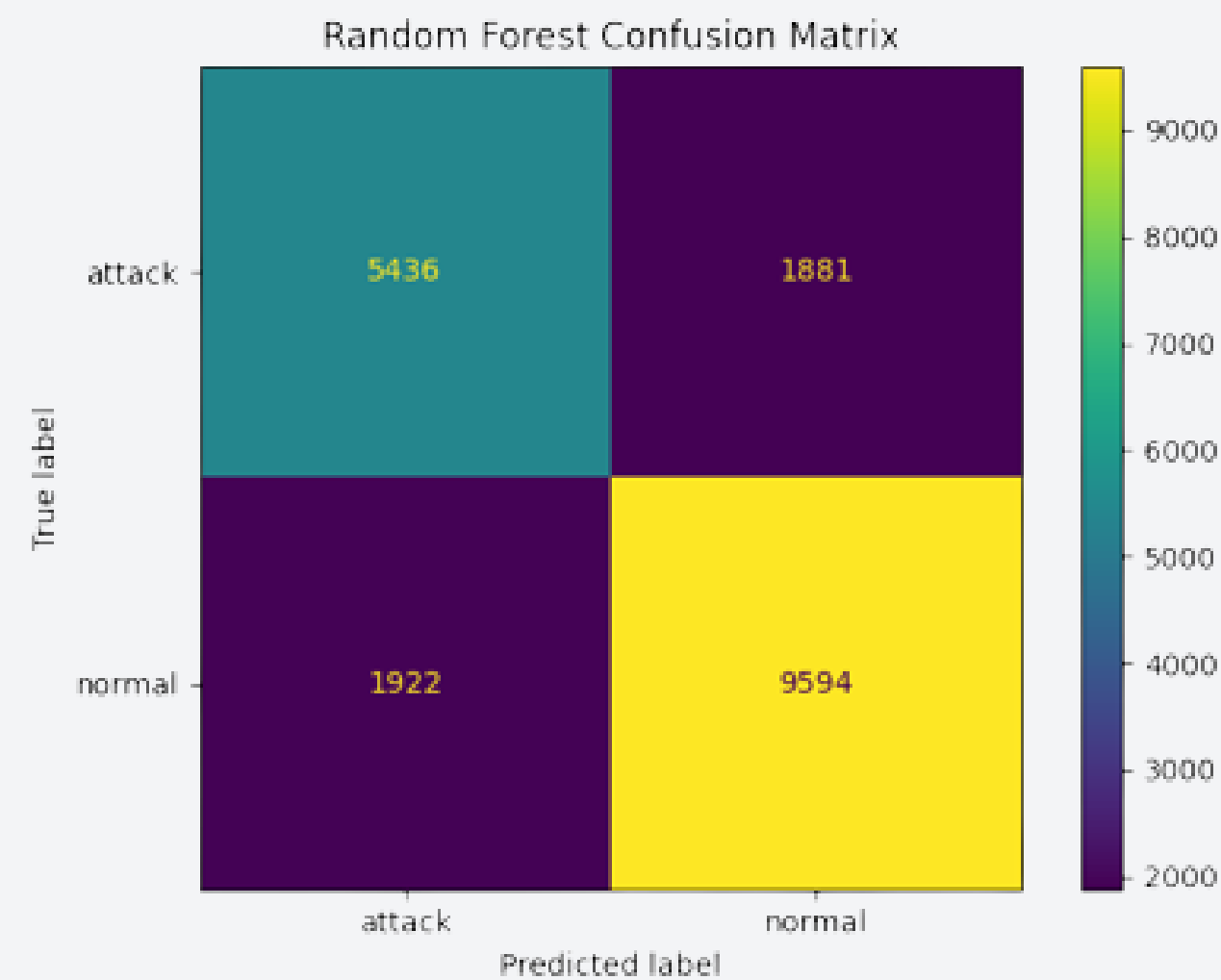
ROC AUC: 0.8975585210002848
```

- Precision — "Of everything I flagged as attack, how much was actually attack?"
- Recall — "Of everything that was actually attack, how much did I catch?"
- F1-score — "Overall, how well balanced are precision and recall together?"

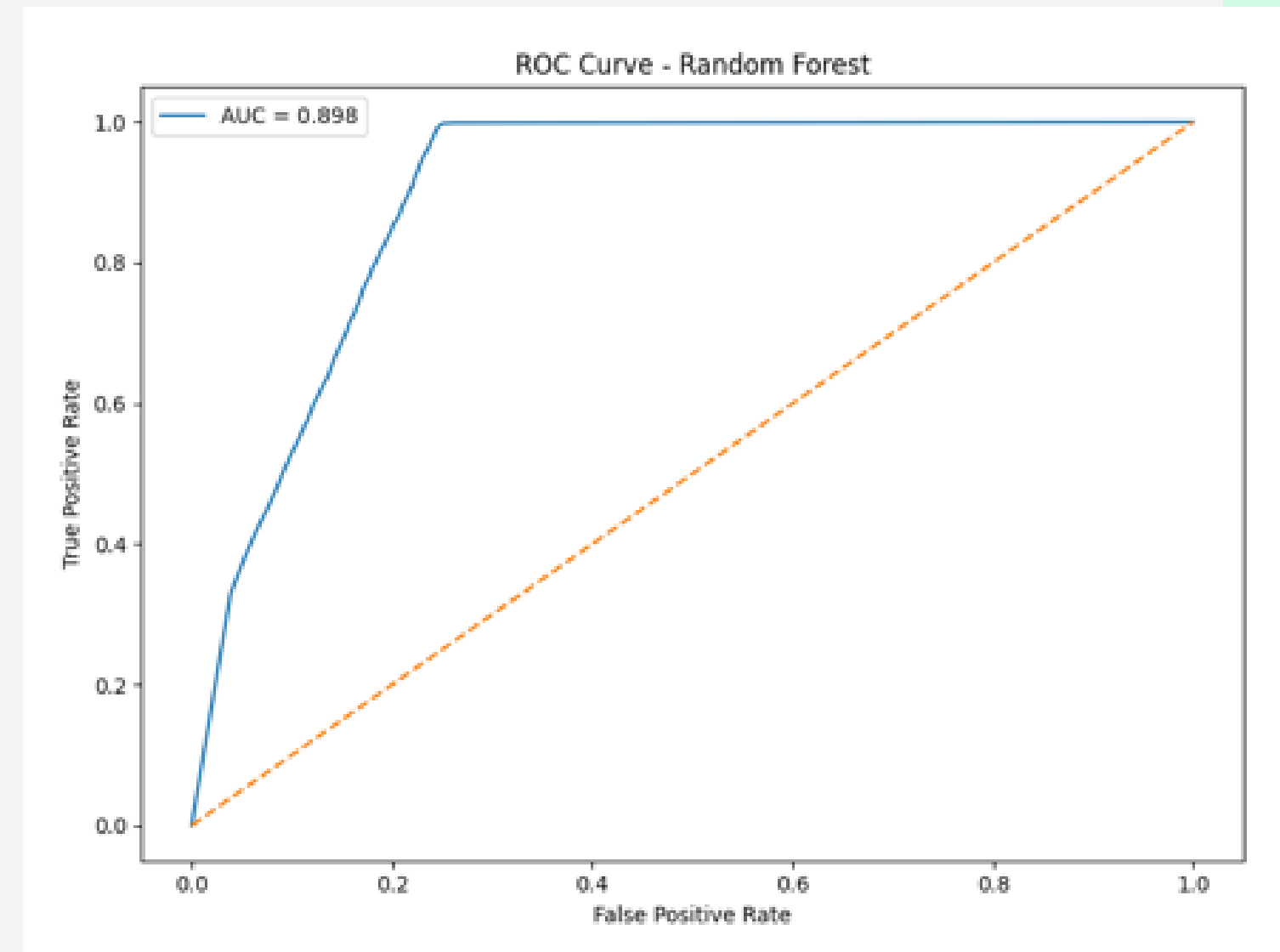


- True Positive - attack, detect as attack
- False Negative - attack, detect as normal
- False Positive - normal, detect as attack
- True Negative - normal, detect as normal

True Positive, False Negative
False Positive, True Negative



True Positive, False Negative
False Positive, True Negative



How good is the model at telling attacks and
normal logs apart

ROC-AUC: 0.898 (Good)

Feature	Importance	Type
anomaly_score	45.10%	numeric
firewall	4.80%	text (TF-IDF word)
severity_score	4.20%	numeric
low	3.90%	text
iot	3.40%	text
device_type_firewall	3.30%	categorical (one-hot)
source_type_iot	3.20%	categorical
traffic	3.10%	text
block	2.60%	text
network	2.50%	text
deny	2.40%	text
high	2.20%	text

Random Forest Feature Importance

- A feature's importance score reflects how much that feature helped the trees make correct splits, averaged across all 150 trees
- The higher the number, the more the model relied on that feature to make its decisions

Demonstration



Conclusion



The proposed AI-based framework successfully unified logs from Firewall, Android, and IoT sources for SIEM monitoring. By using NLP and Random Forest, the system supports automated attack classification and risk scoring, helping to improve security monitoring and analysis.

Thank You

End of Presentation

